Module 9: Outlier Detection

- Basic concepts
- Statistical approaches
- Proximity-based approaches
- Reconstruction-based approaches
  - Matrix factorization-based methods for numeric data
  - Pattern-based compression methods for categorical data
- Clustering and classification-based approaches
- Mining contextual and collective outliers
- Outlier detection in high-dimensional data

Reconstruction-Based Approaches: Example

- Data mining: KDD and ICDM
- Software engineering: FSE and ICSE

Researchers	Publication Venues (Original)
John	KDD, ICDM
Tom	KDD, ICDM
Bob	KDD, ICDM
Carl	ICDM, FSE, ICSE
Van	FSE, ICSE
Roy	FSE, ICSE

(a) Original representation

Researchers	Publication Areas (Succinct)
John	Data Mining
Tom	Data Mining
Bob	Data Mining
Carl	Software Engineering
Van	Software Engineering
Roy	Software Engineering

(b) Succinct representation

Key questions in Reconstruction-based Outlier Detection Methods

- How to find the succinct representation
- How to use the succinct representation to reconstruct the original data samples
- How to measure the quality (i.e., goodness) of reconstruction

Matrix Factorization Based Methods for Numeric Data

- Each data sample is represented as an attribute vector
- The entire input data set is then represented as a data matrix
  - Rows are different data samples
  - Columns are different attributes
  - Entries of the data matrix indicate the attribute values of the corresponding data samples
- Matrix factorization: approximate the data matrix by two or more low-rank matrices
- The reconstruction error of each data sample is used as an indicator of the outlierness: the higher the reconstruction error, the more likely the given data sample is an outlier

Illustration

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\ \text{Author}}_{\text{Reconstructed Data Matrix } \hat{\boldsymbol{X}}}
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Algorithm

Algorithm: Matrix factorization based outlier detection

Input:

- $X_i = (X_{i,1}, ..., X_{i,m})$ $(i = 1, ..., n)$, n data samples each of which is represented by an m dimensional numerical attribute vector;
- r, the rank;
- k, the number of outliers.

Output: A set of k outliers.

Method:

//each row of $\boldsymbol{X}$ is a data sample
(1) Represent the input data samples as an $n \times m$ data matrix $\boldsymbol{X}$;
(2) Approximate the data matrix by two rank r-matrices: $\boldsymbol{X} \approx \boldsymbol{F}\boldsymbol{G}$;
(3) **for** $(i = 1, ..., n)$\{ // for each sample
//compute the reconstruction error
//$\boldsymbol{F}(i,j)$ is the element of $\boldsymbol{F}$ at the i^{th} row and the j^{th} column
//$\boldsymbol{G}(j,:)$ is the j^{th} row of $\boldsymbol{G}$
(4) Compute $r_i = \|X_i - \hat{X}_i\|^2 = \|X_i - \sum_{j=1}^{r} \boldsymbol{F}(i,j)\boldsymbol{G}(j,:)\|^2$;
\}
(5) Return k samples with the largest reconstruction errors r_i $(i = 1, ..., n)$.

Singular Value Decomposition (SVD)

- Given an input data matrix X, SVD approximates it by the multiplication of three rank-r matrices X ≈ UΣV’
- U and V are orthonormal matrices, and their columns are called left singular vectors and right singular vectors of data matrix X, respectively
- Σ is a diagonal with non-negative diagonal elements called singular values